

Problem

Given N floating-point numbers in the range [0, 1), sort them using the Bucket Sort algorithm.

Input Format

N a1 a2 a3 ... aN

Constraints

$1 \leq N \leq 100000$ $0 \leq a_i < 1$

Output Format

Print the sorted array rounded to 2 decimal places.

Sample Input 0

6
0.90 0.80 0.70 0.60 0.50 0.40

Sample Output 0

0.40 0.50 0.60 0.70 0.80 0.90

Sample Input 1

6
10 20 40 50 30 60

Sample Output 1

10 20 30 40 50 60

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Language Python 3



```
1 # Enter your code here. Read input from STDIN.
2 n = int(input())
3 arr = list(map(float, input().split()))
4 buckets = [[] for _ in range(n)]
5 minimum = min(arr)
6 maximum = max(arr)
7 for num in arr:
8     if maximum == minimum:
9         index = 0
10    else:
11        index = int((num - minimum) / (maximum
12        buckets[index].append(num)
13 for bucket in buckets:
14     bucket.sort()
15 for bucket in buckets:
16     for num in bucket:
17         if num.is_integer():
18             print(int(num), end=" ")
19         else:
20             print(f"{num:.2f}", end=" ")
21
22
```

Line: 14 Col: 18

Upload Code as File

Run Code

Submit Code

☐ Test against custom input

Congratulations

You solved this challenge. Would you like to challenge your friends?

Test case 0

Test case 1

Compiler Message

Success